

Image Processing Fundamentals Reflection Journal

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Section 1: Technical Understanding

The most surprising discovery in this lab was that a digital image is not stored as one continuous picture. It is represented as a matrix of numbers. A grayscale image uses a two-dimensional matrix, while a color image uses a three-dimensional array containing red, green, and blue values. Looking at the separate channels made it clear that each visible color comes from a combination of numerical intensities. This changed the way I think about images because every edit can be traced back to changes in those values.

The mathematical operations in the notebook had direct visual effects. Adding a constant increased brightness, multiplying pixel values changed contrast, and convolution used a small kernel to combine neighboring pixels. A blur kernel reduced local differences, a sharpening kernel emphasized differences, and edge-detection kernels responded strongly where brightness changed quickly. Histogram equalization redistributed intensity values, while CLAHE improved local contrast without treating the entire image exactly the same. The most challenging technique was convolution because the result depends on both the values inside a kernel and where those values are placed. A small change in the kernel can create a completely different visual result.

Section 2: Connections and Applications

The lab connects to the Nano Banana demonstration because both traditional image processing and modern AI tools operate on numerical image data. In the lab, filters were designed manually and applied in a known order. In an AI system, many useful filters are learned from training data and combined across multiple layers. Edge detection, texture extraction, color adjustment, and geometric transformations are still important foundations. The main difference is that an AI model learns which combinations are useful instead of relying only on hand-written rules.

These techniques have many real-world uses. Medical imaging systems can use contrast enhancement to reveal low-contrast structures, and edge detection can help outline organs or possible abnormalities. Autonomous vehicles use image processing to find lane boundaries and prepare camera images for object-detection models. Document scanners use grayscale conversion, thresholding, and geometric correction to improve readability. Security cameras use noise reduction and enhancement in poor lighting. In a future project, I would combine traditional and AI methods by first resizing, normalizing, denoising, and enhancing an image before sending it to a trained model. Traditional preprocessing would provide predictable cleanup, while AI would handle recognition or classification.

My three personal experiments also showed why parameter selection matters. Larger Gaussian kernels created stronger smoothing but removed more detail. Lower Canny thresholds detected more edges, including weak details and noise, while higher thresholds produced a cleaner but less complete result. Increasing the CLAHE clip limit strengthened local contrast, but the strongest setting could make texture look unnatural. These comparisons showed that there is rarely one perfect setting for every image.

Section 3: Personal Reflection

The area of image processing that interests me most is medical image enhancement. Small details in an X-ray, CT scan, or MRI can be important, so I would like to learn how enhancement, denoising, segmentation, and AI can work together without creating misleading artifacts. This lab changed my understanding of digital photography and image editing because filters no longer seem like unexplained effects. They are organized mathematical transformations applied to arrays of pixel values.

I still have questions about how professionals choose the best settings automatically and how systems are tested across different cameras, scanners, and patient populations. I also want to understand how AI models explain which image regions influenced a decision. Overall, the lab helped me connect basic mathematics, traditional image-processing tools, and modern AI. It showed that advanced systems are built from the same fundamental ideas: pixels, matrices, filters, transformations, and carefully selected parameters.